

2.1 (a)

xyz	$x+y+z$	$(x+y+z)'$	x'	y'	z'	$x'y'z'$	xyz	(xyz)	$(xyz)'$	x'	y'	z'	$x'y'z'$
000	0	1	1	1	1	1	000	0	1	1	1	1	1
001	1	0	1	1	0	0	001	0	1	1	1	0	1
010	1	0	1	0	1	0	010	0	1	1	0	1	1
011	1	0	1	0	0	0	011	0	1	1	0	0	1
100	1	0	0	1	1	0	100	0	1	0	1	1	1
101	1	0	0	1	0	0	101	0	1	0	1	0	1
110	1	0	0	0	1	0	110	0	1	0	0	1	1
111	1	0	0	0	0	0	111	1	0	0	0	0	0

(b)

xyz	$x+yz$	$(x+y)$	$(x+z)$	$(x+y)(x+z)$
000	0	0	0	0
001	0	0	1	0
010	0	1	0	0
011	1	1	1	1
100	1	1	1	1
101	1	1	1	1
110	1	1	1	1
111	1	1	1	1

(c)

xyz	$x(y+z)$	xy	xz	$xy+xz$
000	0	0	0	0
001	0	0	0	0
010	0	0	0	0
011	0	0	0	0
100	0	0	0	0
101	1	0	1	1
110	1	1	0	1
111	1	1	1	1

(c)

xyz	x	$y+z$	$x+(y+z)$	$(x+y)$	$(x+y)+z$
000	0	0	0	0	0
001	0	1	1	0	1
010	0	1	1	1	1
011	0	1	1	1	1
100	1	0	1	1	1
101	1	1	1	1	1
110	1	1	1	1	1
111	1	1	1	1	1

(d)

xyz	yz	$x(yz)$	xy	$(xy)z$
000	0	0	0	0
001	0	0	0	0
010	0	0	0	0
011	1	0	0	0
100	0	0	0	0
101	0	0	0	0
110	0	0	1	0
111	1	1	1	1

2.2

- (a) = x
- (b) = x
- (c) = y
- (d) = 0
- (e) = $xz + yz + x'y'z'$

2.3

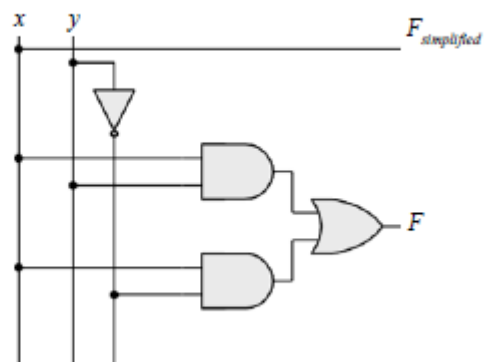
- (a) = y
- (b) = $z(x + y)$
- (c) = $x'y'$

- $(d) = x(w + y)$
- $(e) = 0$
- $(f) = z' + x'y'$

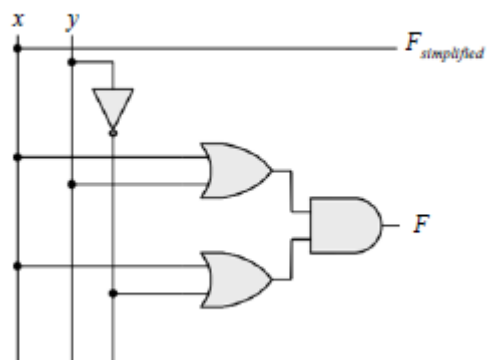
- 2.4
- (a) $= y + z'$
 - (b) $= y(x' + z)$
 - (c) $= x'$
 - (d) $= xy(w + z)$
 - (e) $= wy'$

2.5

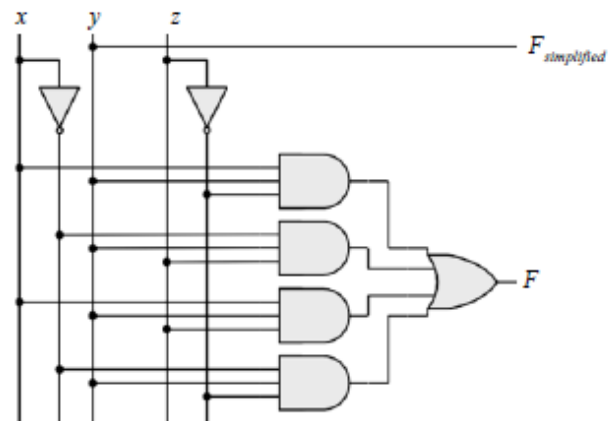
(a)



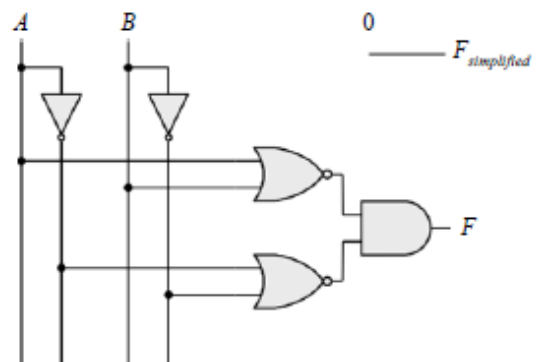
(b)



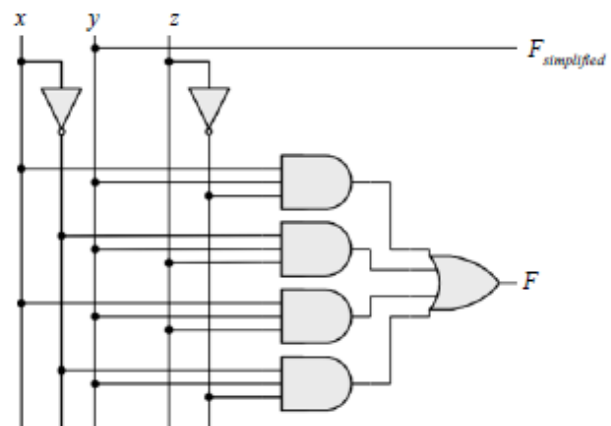
(c)



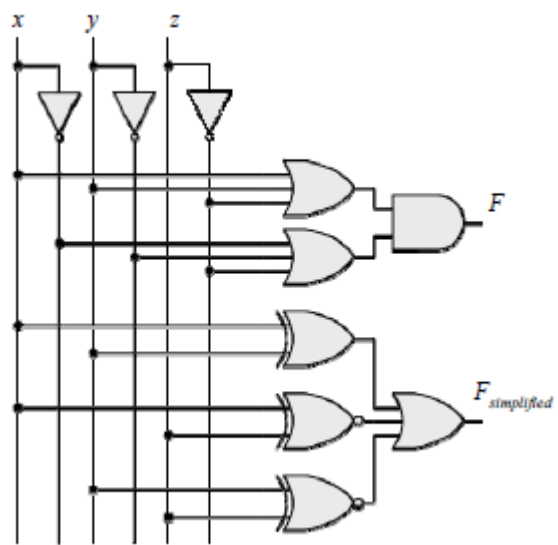
(d)



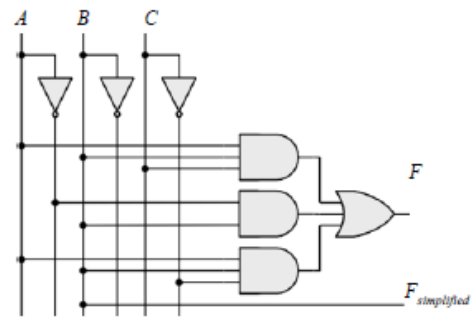
(e)



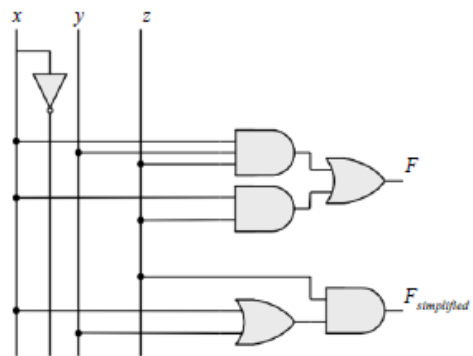
(f)



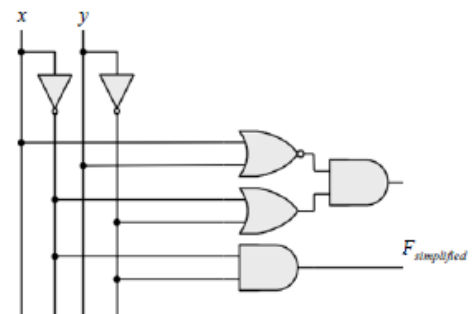
2.6 (a)



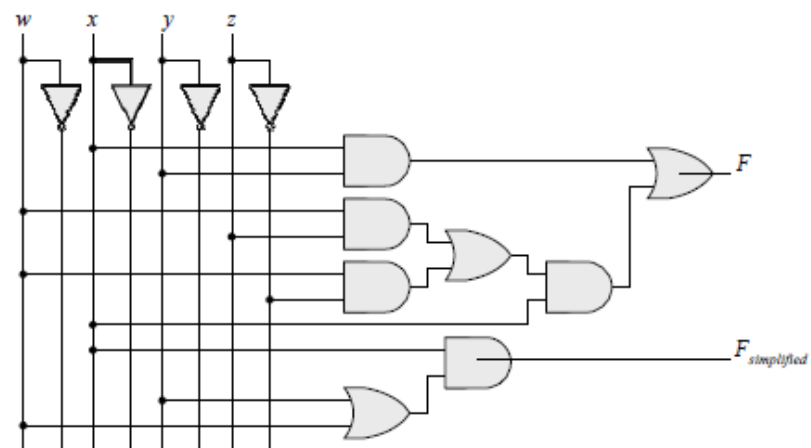
(b)



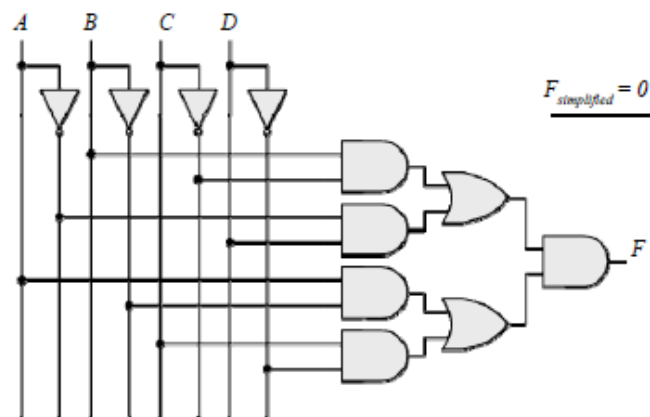
(c)



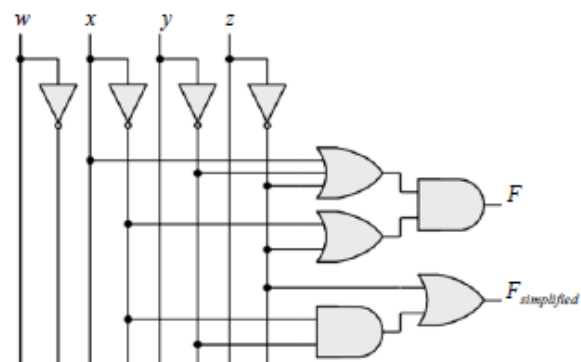
(d)



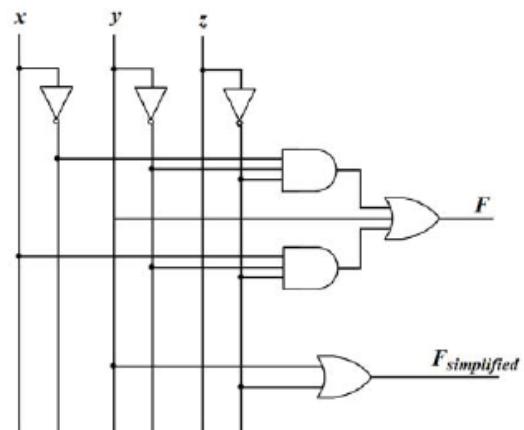
(e)



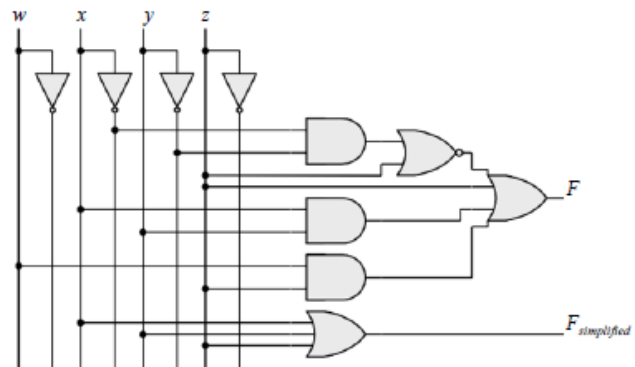
(f)



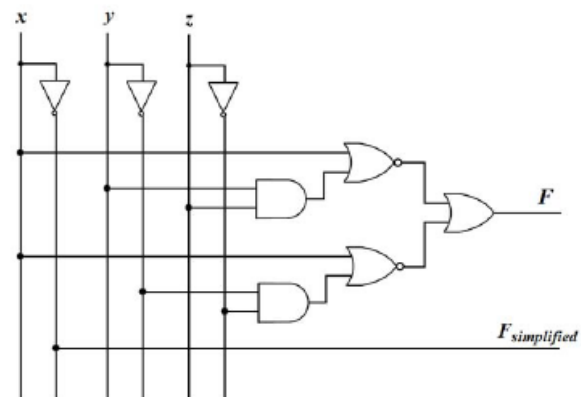
2.7 (a)



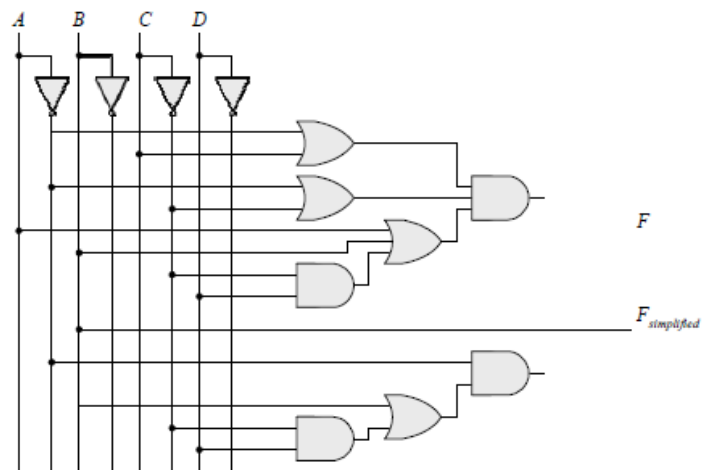
(b)



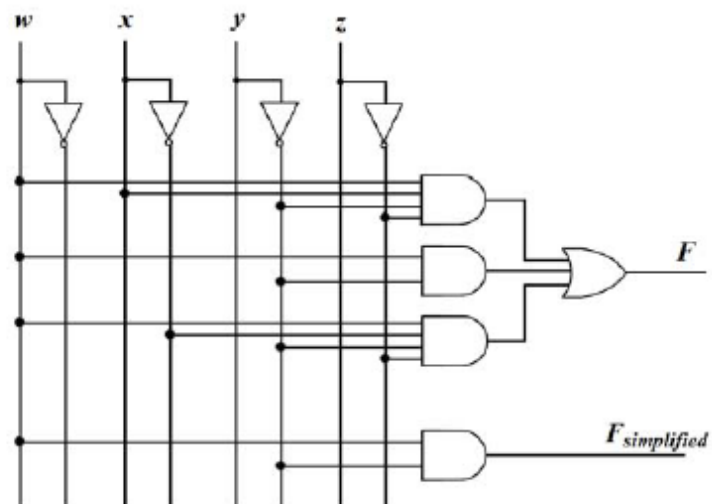
(c)



(d)



(e)



$$\begin{aligned}
2.8 \quad & F = x'y + yz' \\
& F' = xz + y' \\
& FF' = 0 \\
& F + F' = 1
\end{aligned}$$

2.9

$$(a) \quad F' = (x' + y' + z')(x + y + z)$$

$$(b) \quad F' = x'y + y'z + z'x$$

$$(c) \quad F' = (w + x)(w' + x'y'z')$$

$$2.10 \quad (a) \quad F_1 + F_2 = \sum m_{1i} + \sum m_{2i} = \sum (m_{1i} + m_{2i})$$

$$(b) \quad F_1 \cdot F_2 = \sum m_i \sum m_j \text{ where } m_i m_j = 0 \text{ if } i \neq j \text{ and } m_i m_j = 1 \text{ if } i = j$$

$$2.11 \quad (a) \quad F = x'z' + xy + yz = x'y'z' + x'yz' + xyz' + xyz + x'yz = \sum(0, 2, 3, 6, 7)$$

Truth Table of $F = x'z' + xy + yz$:

x	y	z	F
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

- (b) $F = a'b'c' + a'bc + ab'c' + abc = \Sigma(0, 3, 4, 7)$
 Truth Table of $F = a'b'c' + a'bc + ab'c' + abc$:

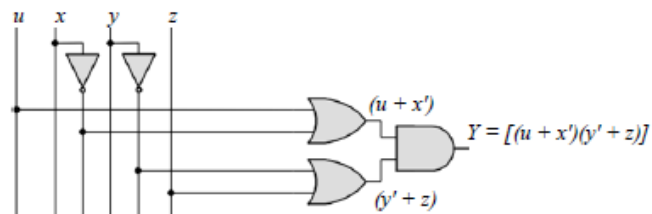
a	b	c	F
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

2.12 $A = 1100_1010$

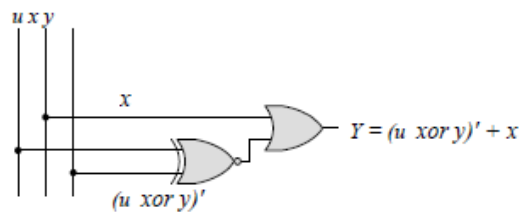
$B = 1001_0011$

- (a) $A \text{ AND } B = 1000_0010$
 (b) $A \text{ OR } B = 1101_1011$
 (c) $A \text{ XOR } B = 0101_1001$
 (d) $\text{NOT } A = 0011_0101$
 (e) $\text{NOT } B = 0110_1100$

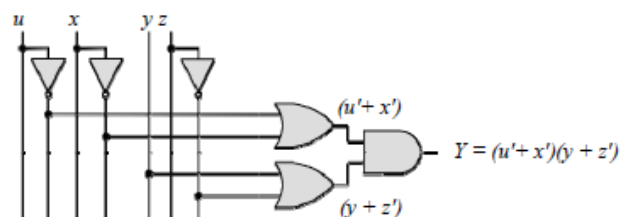
2.13 (a)



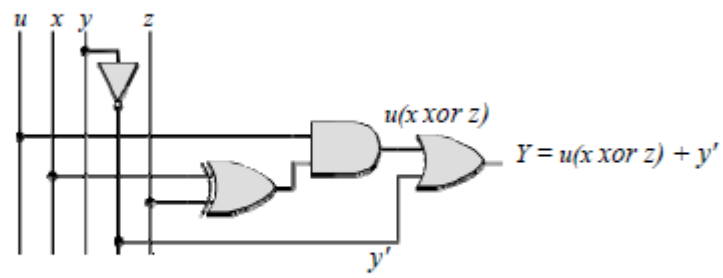
(b)



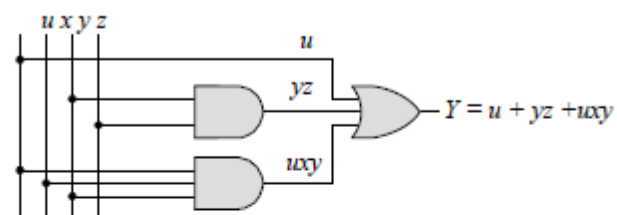
(c)



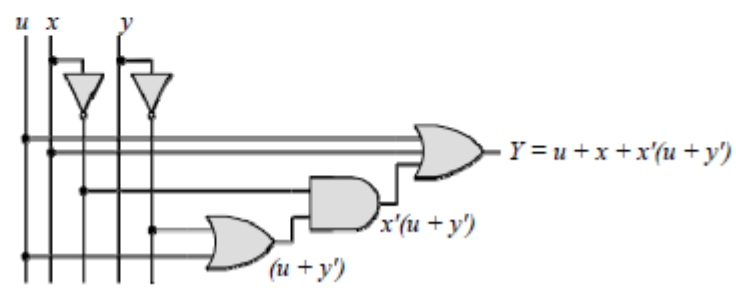
(d)



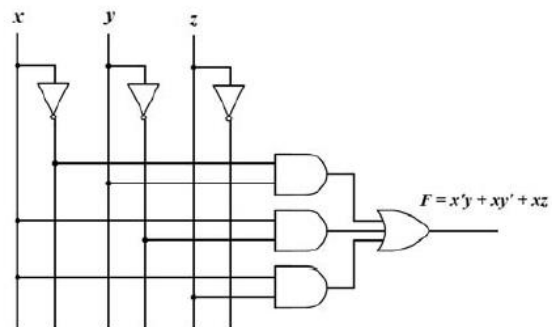
(e)



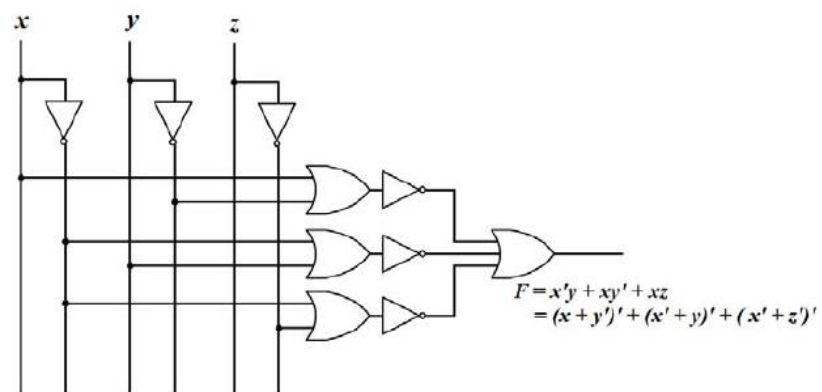
(f)



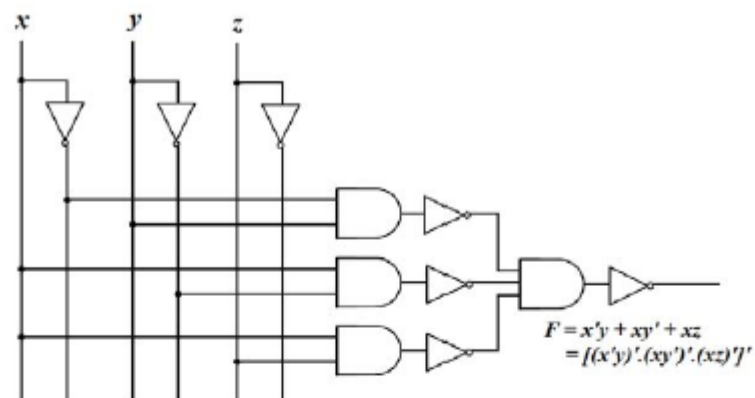
2.14 (a)



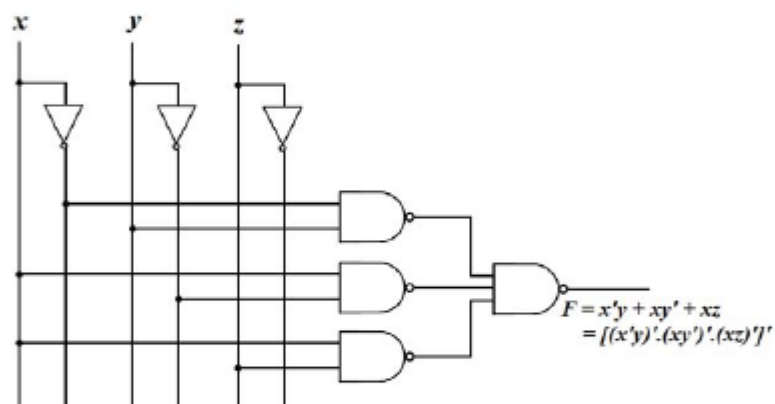
(b)



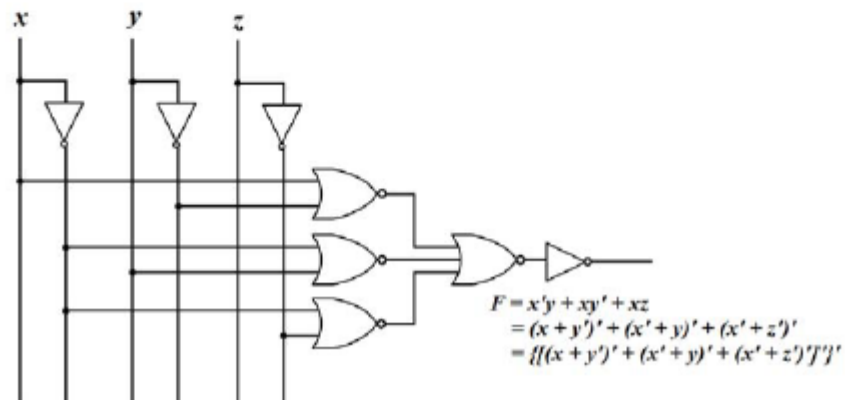
(c)



(d)



(e)



2.15

$$T_1 = A'C + AC + A'BC'$$

$$T_2 = A'C' + AC$$

2.16

(a) 1

(b) $F = (x_n + x'_n)G = 1.$

2.17 (a) $F = (ac + b)(ab + d) = abc + acd + ab + bd = \Sigma(5, 7, 11, 12, 13, 14, 15)$
 $F' = \Sigma(0, 1, 2, 3, 4, 6, 8, 9, 10)$
 $F = \Pi(0, 1, 2, 3, 4, 6, 8, 9, 10)$

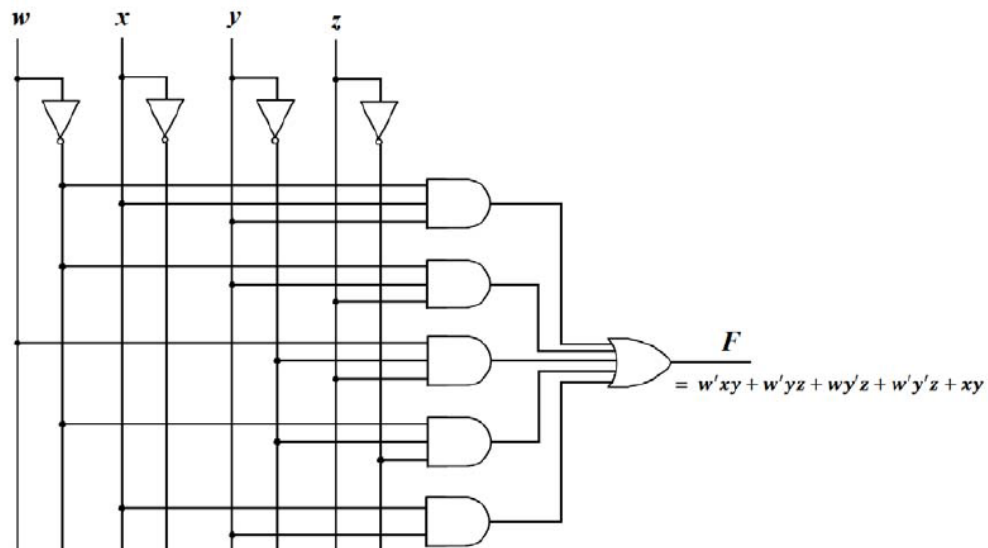
(b) $F = (a' + c' + d')(ab + cd) = a'cd + abc' + abd'$
 $= a'b'cd + a'bcd + abc'd' + abc'd + abc'd' + abcd'$
 $= \Sigma(3, 7, 12, 13, 14)$
 $F' = \Sigma(0, 1, 2, 4, 5, 6, 8, 9, 10, 11, 15)$
 $F = \Pi(0, 1, 2, 4, 5, 6, 8, 9, 10, 11, 15)$

(c) $F = (b + c'd')(a + bc') = ab + bc' + ac'd' + bc'd'$
 $= abc'd' + abc'd + abcd' + abcd + a'bc'd' + a'bc'd$
 $+ abc'd' + abc'd + ab'c'd' + abcd + a'bcd + abcd$
 $= \Sigma(4, 5, 7, 8, 12, 13, 14, 15)$
 $F' = \Sigma(0, 1, 2, 3, 6, 9, 10, 11)$
 $F = \Pi(0, 1, 2, 3, 6, 9, 10, 11)$

(d) $F = a'b'c' + acd + ab'd' + b'cd$
 $= a'b'c'd' + a'b'c'd + ab'cd + abcd + ab'c'd' + ab'cd' + a'b'cd + ab'cd$
 $= \Sigma(0, 1, 3, 8, 10, 11, 15)$
 $F' = \Sigma(2, 4, 5, 6, 7, 9, 12, 13, 14)$
 $F = \Pi(2, 4, 5, 6, 7, 9, 12, 13, 14)$

2.18 (a) $F = w'xy + w'yz + wy'z + w'y'z + xy$
 $F = \Sigma(1, 3, 5, 6, 7, 9, 13, 14, 15)$

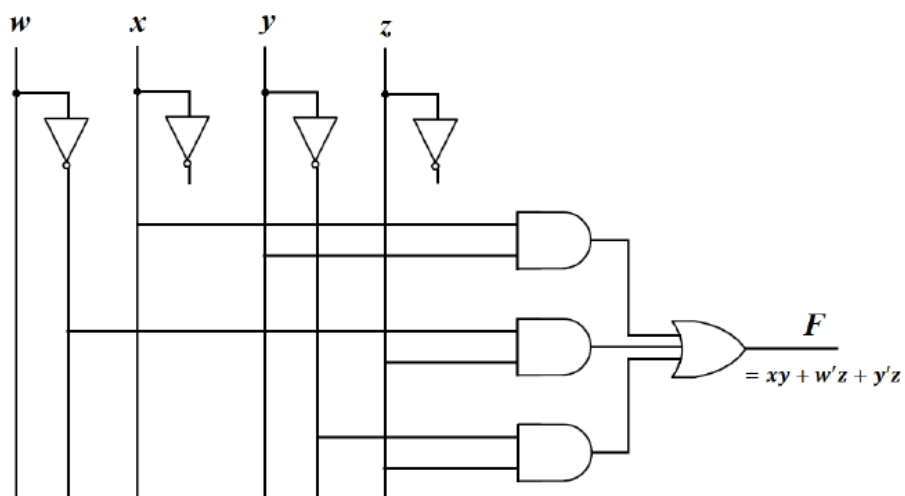
- (b) 5 - Three-input AND gates
1 - Five-input OR gate
3 - Inverters



(c) $F = xy + w'z + y'z$

- (d) $F = xy + w'z + y'z = \Sigma(6, 7, 14, 15, 1, 3, 5, 7, 1, 5, 9, 13) = \Sigma(1, 3, 5, 6, 7, 9, 13, 14, 15)$
[Insert same table as in (a)]

- (e) 2 – Inverters, 3 – Two-input AND gates, 1 – Three-input OR gate



$$2.19 \quad F(A, B, C, D) = AB' + BC + ACD'$$

$$F = \Sigma(6, 7, 8, 9, 10, 11, 14, 15) = \Pi(0, 1, 2, 3, 4, 5, 12, 13)$$

$$2.20 \quad (a) F(w, x, y, z) = \Sigma(1, 5, 7, 11, 12, 14, 15)$$

$$F'(w, x, y, z) = \Sigma(0, 2, 3, 4, 6, 8, 9, 10, 13)$$

$$(b) F(x, y, z) = \Pi(2, 4, 5)$$

$$F'(x, y, z) = \Pi(0, 1, 3, 6, 7) = \Sigma(2, 4, 5)$$

$$2.21 \quad (a) F(x, y, z) = \Sigma(2, 3, 5, 7) = \Pi(0, 1, 4, 6)$$

$$(b) F(A, B, C, D) = \Pi(1, 2, 3, 5, 8, 13) = \Sigma(0, 4, 6, 7, 9, 10, 11, 12, 14, 15)$$

$$2.22 \quad (a) \text{ SOP Form:}$$

$$(w + xy')(x + y'z) = wx + wy'z + xy'z + xy'z = wx + wy'z + xy'z$$

POS Form:

$$(w + xy')(x + yy'z) = (w + x)(w + yy')(x + yy')(x + z)$$

$$(b) \text{ SOP Form:}$$

$$x'y' + (w' + y'z')(z' + x'y')$$

$$= x'y' + x'z' + w'x'y' + y'z' + x'y'z'$$

$$= x'y'(1 + w') + w'z' + (1 + x')y'z'$$

$$= x'y' + w'z' + y'z'$$

POS Form:

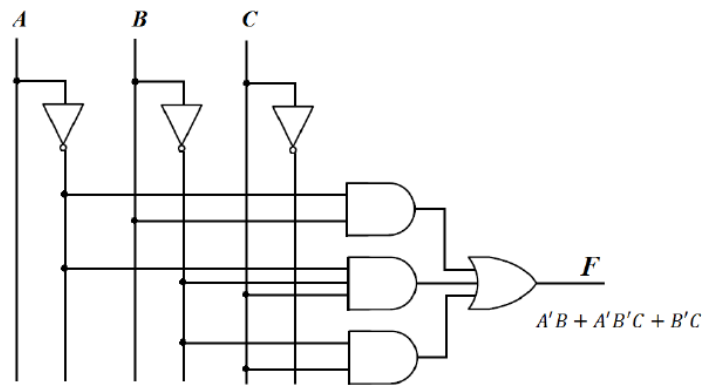
$$x'y' + (w' + y'z')(z' + x'y')$$

$$= x'y' + w'z' + y'z' = x'y' + z'(w' + y')$$

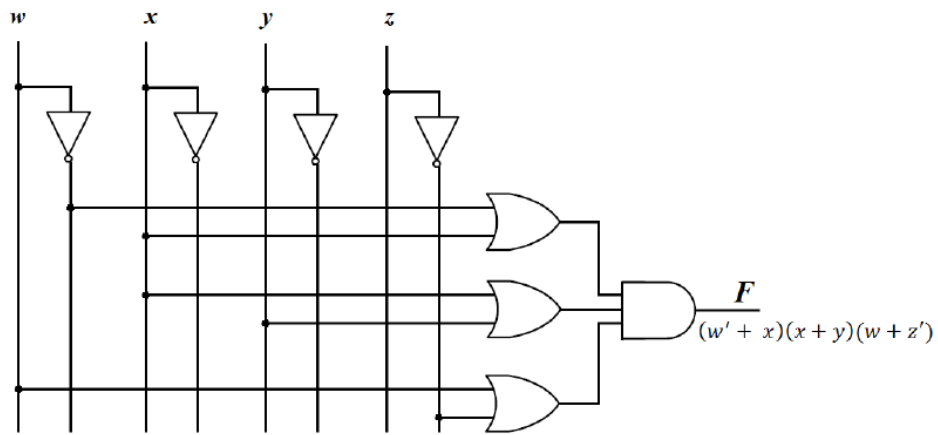
$$= (x'y' + z')(x'y' + w' + y') = (x' + z')(y' + z')(w' + y'(x' + 1))$$

$$= (x' + z')(y' + z')(w' + y')$$

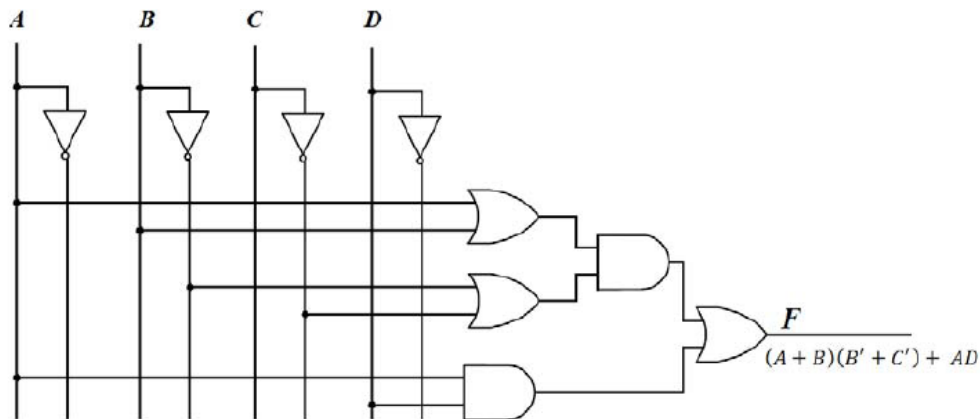
2.23 (a) $A'B + A'B'C + B'C$



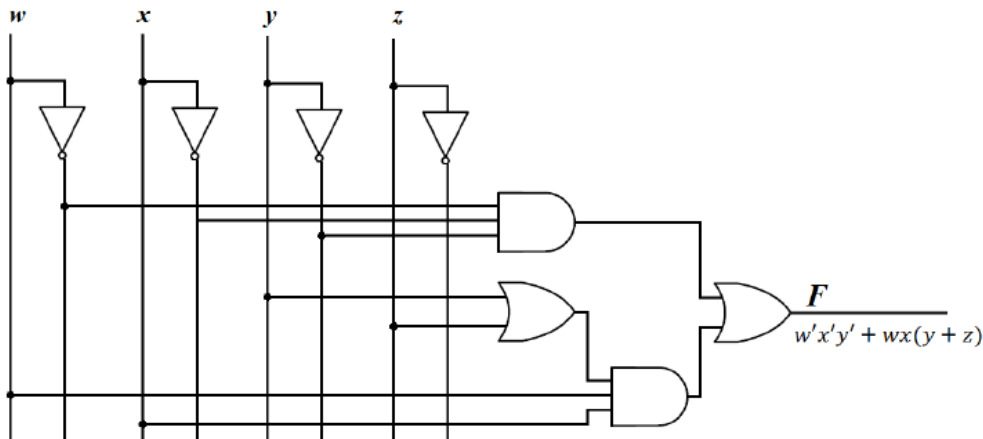
(b) $(w' + x)(x + y)(w + z')$



(c) $(A+B)(B'+C')+AD$



(d) $w'x'y' + wx(y+z)$



2.24 Dual of the expression $x'y + (x+z)(x+y')$ is:

$$(x' + y)(xz + xy')'$$

2.25 (a) $x|y = xy' \neq y|x = x'y$ Not commutative
 $(x|y)|z = xy'z' \neq x|(y|z) = x(yz')' = xy' + xz$ Not associative

(b) $(x \oplus y) = xy' + x'y = y \oplus x = yx' + y'x$ Commutative

$$(x \oplus y) \oplus z = \sum(1, 2, 4, 7) = x \oplus (y \oplus z) \text{ Associative}$$

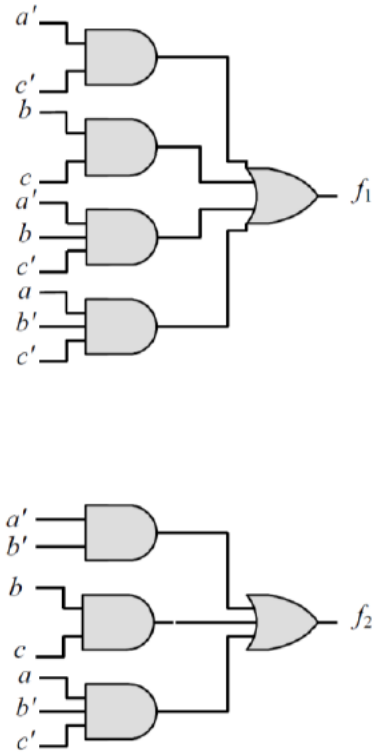
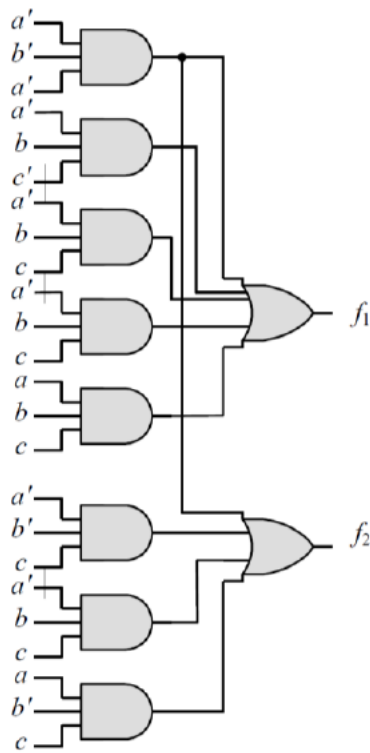
2.26

Gate		NAND (Positive logic)		NOR (Negative logic)	
x y	z	x y	z	x y	z
L L	F	0 0	1	1 1	0
L H	F	0 1	1	1 0	0
H L	F	1 0	1	0 1	0
H H	L	1 1	0	0 0	1

Gate		NOR (Positive logic)		NAND (Negative logic)	
x y	z	x y	z	x y	z
L L	F	0 0	1	1 1	0
L H	L	0 1	0	1 0	1
H L	L	1 0	0	0 1	1
H H	L	1 1	0	0 0	1

2.27 $f_1 = a'b'c' + a'bc' + a'bc + ab'c' + abc = a'c' + bc + a'bc' + ab'c'$

$f_2 = a'b'c' + a'b'c + a'bc + ab'c' + abc = a'b' + bc + ab'c'$



2.28 (a) $y = a(bcd)'e = a(b' + c' + d')e$

(b) $y_1 = a \oplus (c + d + e) = a'(c + d + e) + a(c'd'e') = a'c + a'd + a'e + ac'd'e'$
 $y_2 = b'(c + d + e)f = b'cf + b'df + b'ef$

2.29 L.H.S. = $a'd' + b'c' + c'd'$
 $= (a' + c')d' + b'c' = (b'c' + d')(b'c' + a' + c')$
 $= (b' + d')(c' + d')(a' + c'(b' + 1))$
 $= (a' + c')(b' + d')(c' + d') = \text{R.H.S.}$

Hence the equation is true.

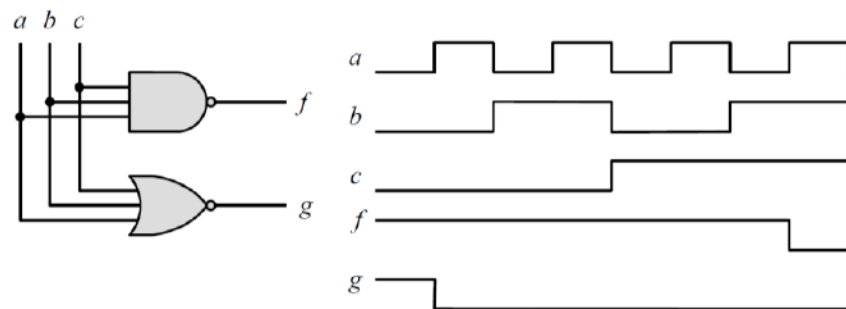
2.30

$$= wx' + wy' + wz' + xy' + xz' + x'y + y'z'$$

2.31

$$= (a' + b)(b + c')(a + b + c')(a + b' + c)(a' + b + c)$$

2.32



2.33

